



# How to Train Your AI Dragon: Email #1 - What AI Actually Is (and Isn't)

Welcome! I am thrilled that you have joined me for this free, quick, short email course on AI. Thank you for subscribing. Starting today and over the next six weeks, you'll receive six insightful emails that will guide you through the fascinating world of Artificial Intelligence. This is email #1: What AI Actually Is (and Isn't).

In the 2010 film "How to Train Your Dragon," the villagers initially misunderstand the nature of dragons, mistaking them for threats. This dynamic mirrors the cautious perspective many social science researchers adopt towards AI. Just as the hero Hiccup learns to approach the dragon differently, researchers can rethink their relationship with AI tools.

So, what exactly is AI? At its core, AI refers to the simulation of human intelligence processes by machines, especially computer systems. This encompasses learning (the acquisition of information and rules for using it), reasoning (the ability to draw conclusions based on rules), and self-correction.

From chatbots to autonomous vehicles, AI is transforming industries and reshaping our day-to-day lives. Unfortunately, the conversation around AI tends to oscillate between utopian hype and existential panic, often leaving researchers uncertain about its practical benefits. Many in academia stand at a distance, grappling with legitimate concerns about plagiarism, data privacy, and the potential homogenization of ideas. Yet, as this series will explore, responsible use of AI can genuinely transform workflows.

### **So what exactly are generative AI (genAI) tools, compared with traditional machine learning (ML) approaches?**

There are lots of different genAI models. Some create images, some can make videos, and some are [pre-trained](#) on huge knowledge bases. The type of genAI we will be focusing on is the Large Language Model, or LLM, of which ChatGPT is one of the better known (there are plenty of others). Under the hood, what LLMs actually do involves next-token prediction and pattern matching over training data. On its own, it is not actually reasoning, thinking, or knowing. In addition, for researchers it's important to note that LLMs like ChatGPT are typically black boxes lacking explainability and reproducibility.

In contrast, analytic and predictive ML tools, such as random forests and natural language processing (NLP) algorithms, generally provide explainable and reproducible results. They are typically established and published packages and programs that can help researchers in data mining, linkage, analysis, and prediction.

Understanding these distinctions is crucial for researchers navigating the evolving landscape of AI. You can learn more about this in my recent blog

post, [Medical Care and AI](#).

Ultimately, genAI doesn't replace the work researchers do; rather, it changes the way they can approach it. Just as Hiccup showed the villagers new possibilities by reframing their fears, researchers can harness AI to enhance their methodology and improve outcomes in their work.

In the next email, we'll delve into our first specific use case: a literature synthesis workflow. See you in a week!

Sincerely,

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